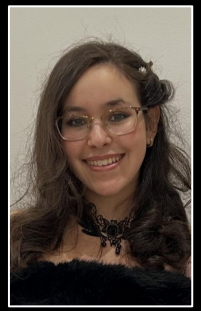
Improving On-Stage Sound Experience

AuricleSync

Client: Mathura Shivakaran

Lina Habib



Naufal As Shodiq



Andre Daminar



Cheah Jing Qian



Chloe Chan



Zhuangkaixuan Liu



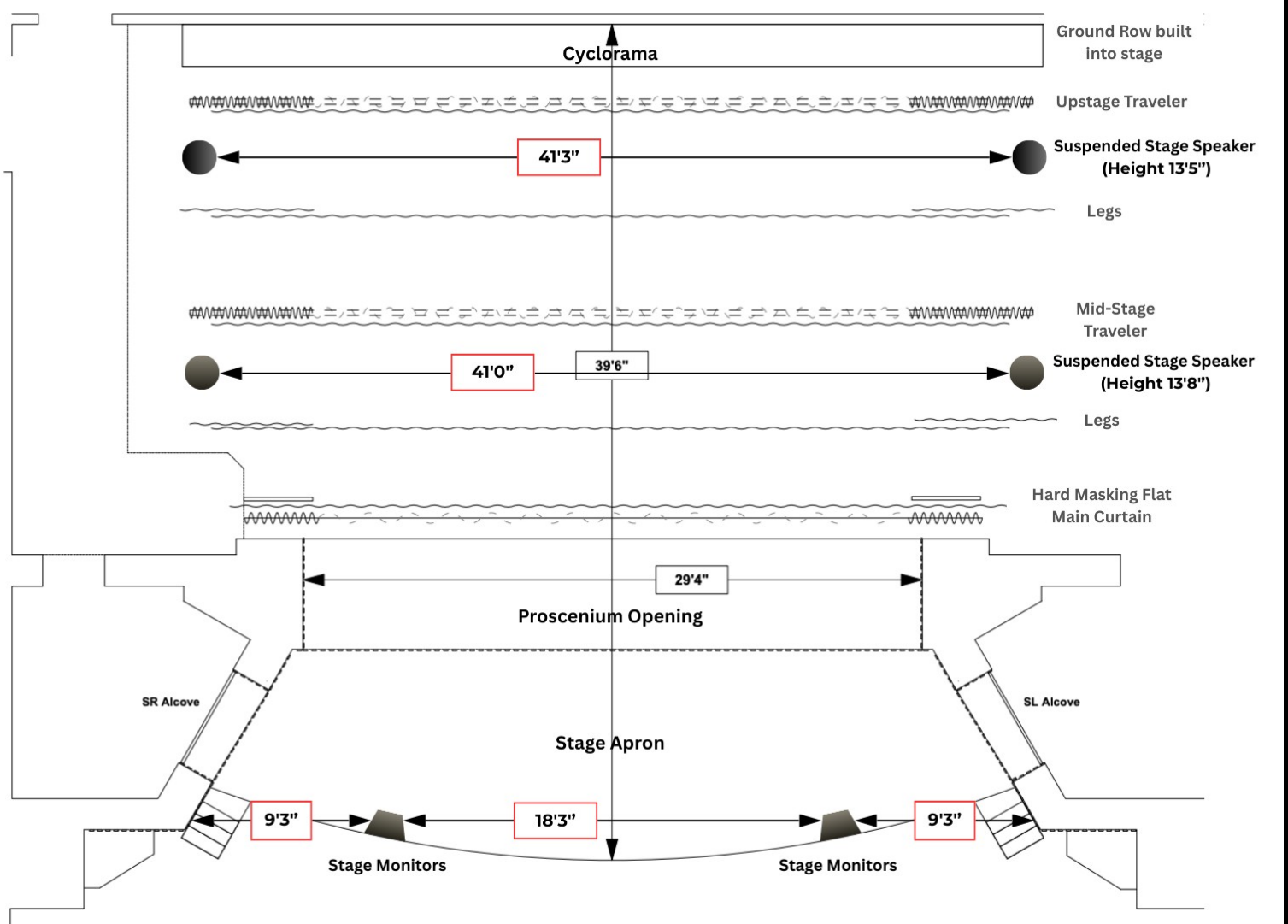
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Cyclorama
Ground Row built into stage
Upstage Traveler
41'3"
Suspended Stage Speaker (Height 13'5")
Legs
Mid-Stage Traveler
41'0"
39'6"
Suspended Stage Speaker (Height 13'8")
Legs
Hard Masking Flat
Main Curtain
29'4"
Proscenium Opening
SR Alcove
SL Alcove
Stage Apron
9'3"
18'3"
9'3"
Stage Monitors
Stage Monitors



Lina Habib

Three criteria guiding the creation of AuricleSync



Lina Habib



CLARITY



ACCURACY



SYNCHRONIZATION





Naufal
As Shodiq

The three criteria guided the creation of AuricleSync

CLARITY

ACCURACY

SYNCHRONIZATION



Naufal
As Shodiq

AuricleSync Performs Better Than Current Hart House Theatre Speaker System by Meeting Key Criteria

Measure of Success: What is the sound latency associated with AuricleSync?



Andre Daminar

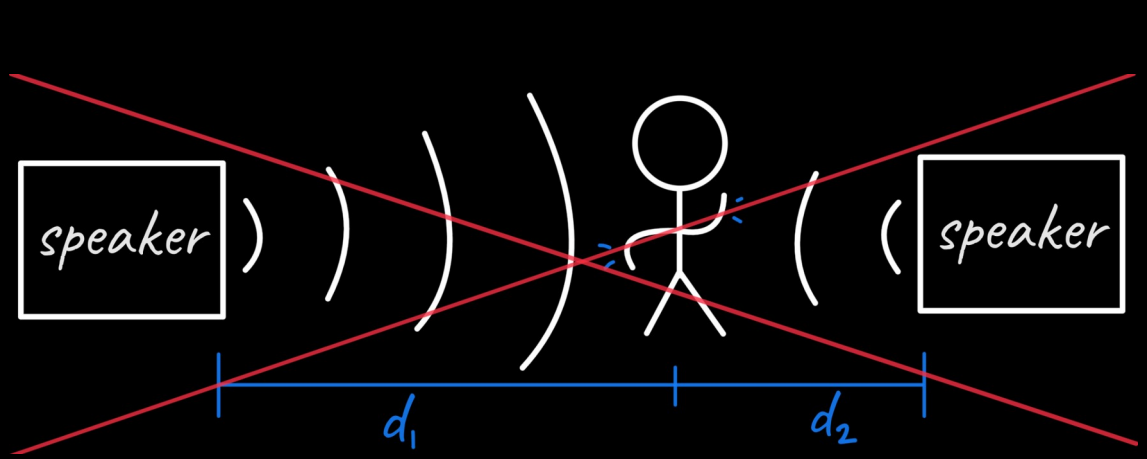
Objective Goal: Minimize audio latency	**Formula Used**
≤10 ms	Time = Distance / Speed of Sound
Speed of Sound	**Distance from outer ear to ear drum**
343 m/s	0.025 m
Calculated measurement	
0.0729ms *(rounded to 3 s.f.)*	

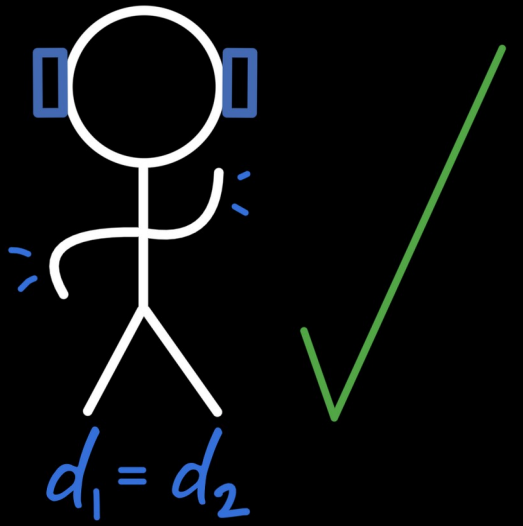
CLARITY ACCURACY SYNC

AuricleSync minimizes distances which guarantees synchronized sound



Andre Daminar



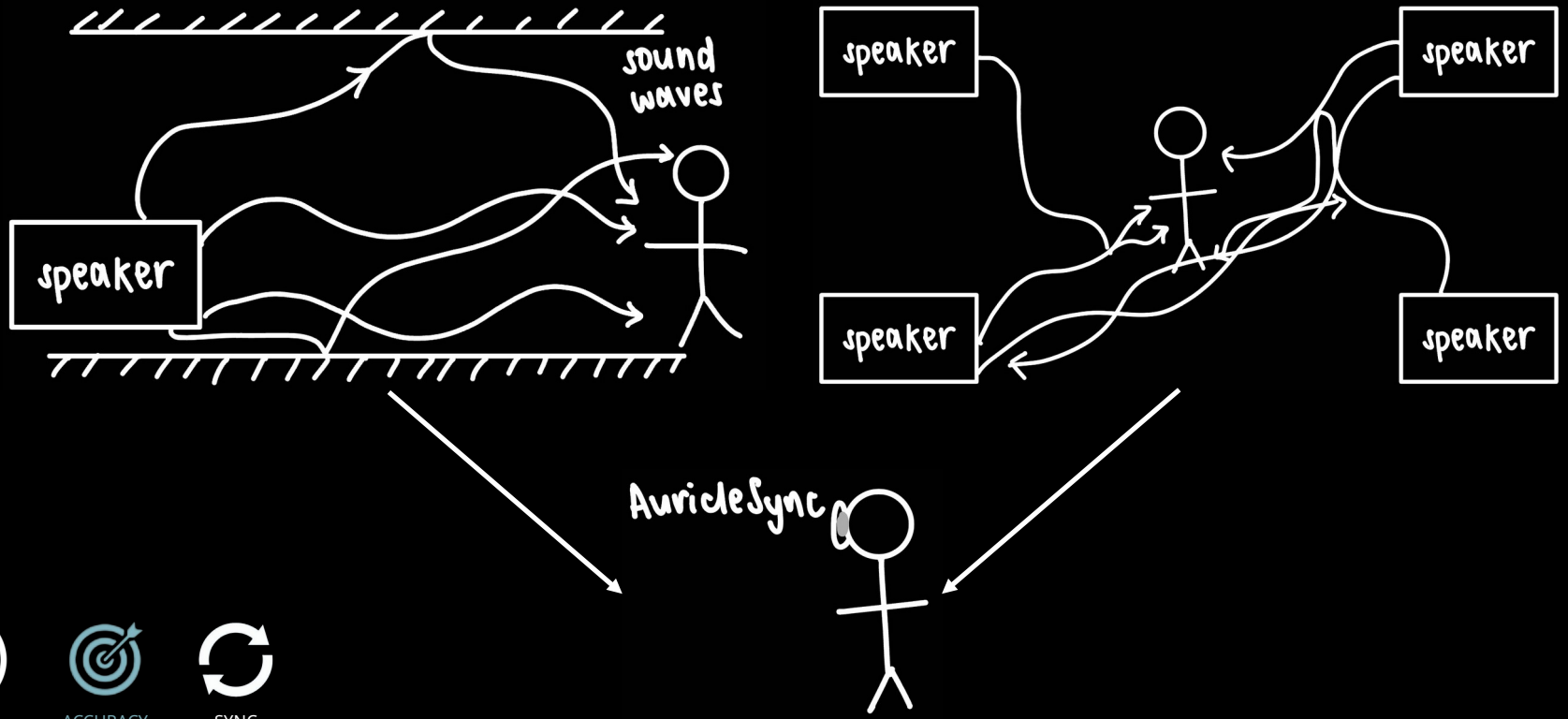


CLARITY ACCURACY SYNC

AuricleSync minimizes sound wave interference, improving sound accuracy

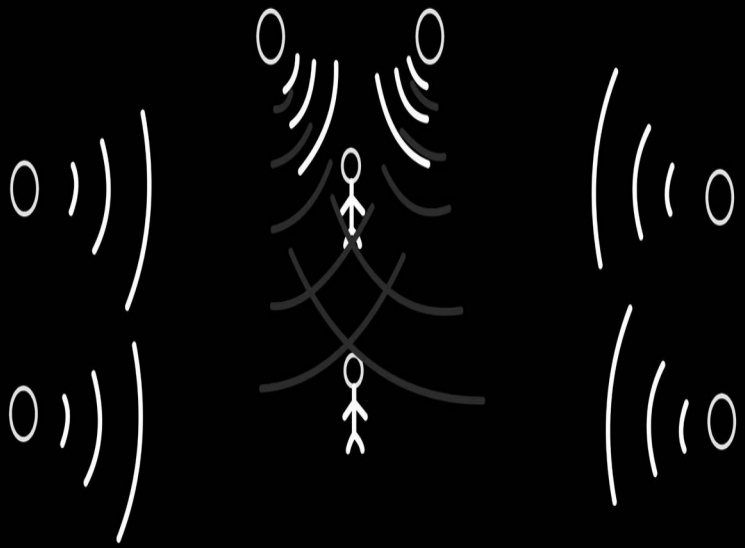


Cheah Jing Qian



CLARITY ACCURACY SYNC

AuricleSync is adjustable to adequately project volume



- Eliminates uneven volume from 6-speaker stage setup
- Individual volume control
- Sound intensity $\propto 1 / r^2$



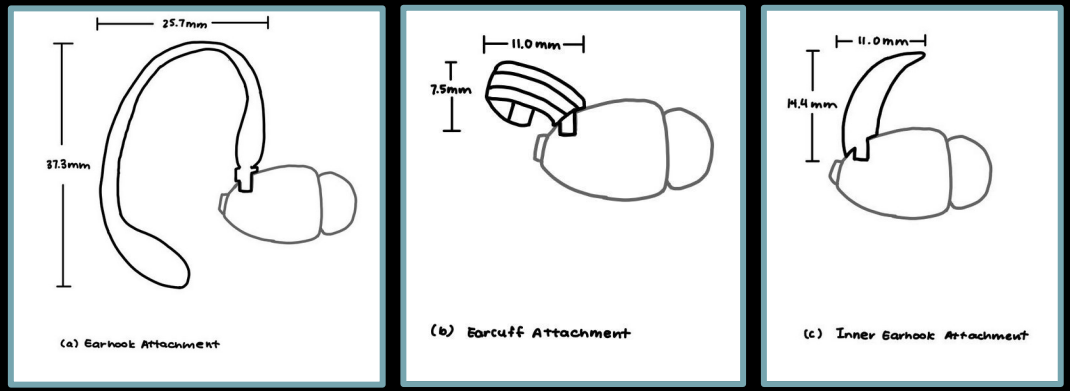
Zhuangkaixuan Liu

CLARITY ACCURACY SYNC

AuricleSync adapts to performer needs while maintaining usability during performance

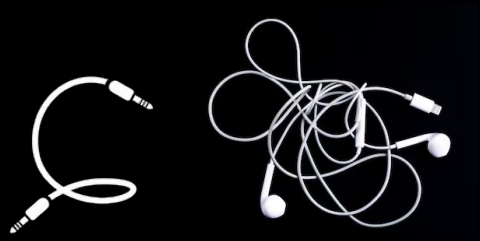


Chloe Chan



Interchangeable attachment options

- Discreet securing attachments
- Minimizes interference with performer motion



CLARITY ACCURACY SYNC EASE OF USE

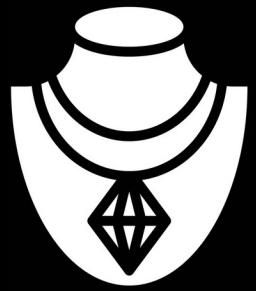
Next steps



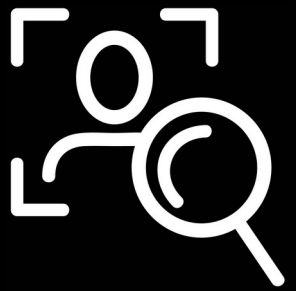
Cheah Jing Qian



Influence of sound from audience speakers with prototype



How prototype can integrate with different costumes



Human testing

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Appendix A: Latency Calculations - Best Case

best case @ "center stage"

time from front speakers: $t = \frac{d}{v}$, $\frac{41'}{2} = 6.223\text{m}$

$$= \frac{\sqrt{\bar{y}^2 + 6.223^2 + 4.1656^2}}{343}$$

$$= 0.022059\text{ s} \quad (5\text{ s.f.})$$

time from back speakers: $t = \frac{d}{v}$, $\frac{41'3"}{2} = 6.265\text{m}$

$$= \frac{\sqrt{\bar{y}^2 + 6.265^2 + 4.0894^2}}{343}$$

$$= 0.022039\text{ s} \quad (5\text{ s.f.})$$

$\Delta t \approx 2 \times 10^{-5}\text{ s}$

$= 20\text{ ms}$

Appendix B: Latency Calculations - Worst Case

Worst case below one of the speakers

time from below the front right speaker

$$t = \frac{d}{v} = \frac{4.1656}{343} \quad \text{— 13'8" in m}$$

$$= 0.012145 \quad (5 \text{ s.f.})$$

time from back left speaker

dist b/w speakers / 1.5" = 0.0381 m / 41' = 12.4968 m

$$t = \frac{d}{v} = \frac{\sqrt{4.1656^2 + 12.5349^2 + 2.159^2}}{343}$$

$$= 0.039021 \text{ s}$$

$$\Delta t \simeq 0.026876 \text{ s}$$

$$= 27 \text{ ms}$$

Appendix C: Design Objectives (Ranked)

Main Objectives	Specified Metric	Goal
Improve Audio Accuracy	Frequency Response (dB)	A flat frequency response (+/- 3dB maximum variation) across the frequency range of 20 Hz to 20 kHz. See Appendix E.2 for a sample graph [21].
Minimize Audio Latency	Milliseconds (ms)	≤10 ms [22].
Volume on Stage	Decibels (dB)	80-85dB [23].
Affordable	$CAD	≤$1000 (Appendix A)
Ease of Use	Steps to operate	≤3 steps [24].

Appendix D: Hart House Frequency Response Test

